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Dolphin: A Python package for large-scale InSAR PS/DS processing

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Software

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Summary

`dolphin` is a Python library for creating maps of land surface displacement using the remote sensing technique called Interferometric Synthetic Aperture Radar (InSAR). Conventional InSAR uses pairs of SAR images to get a single map of the relative displacement between the two acquisition times. `dolphin` uses state-of-the-art multi-temporal algorithms to reduce the impact of noise sources and produce long time series of displacement at fine resolution.

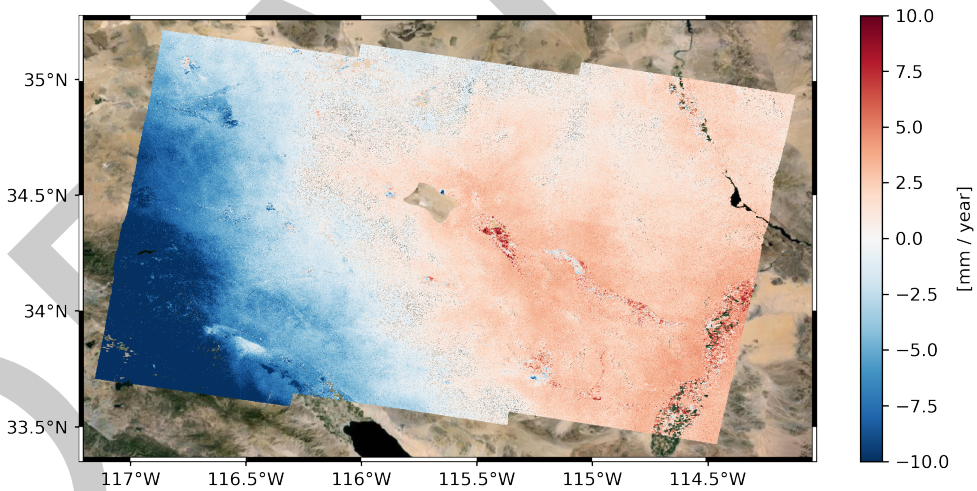

Figure 1: Average surface displacement velocity along the radar line-of-sight between February, 2017 and December, 2020. Red (blue) indicates motion towards (away from) the satellite.

Statement of need

The Sentinel-1 satellite from the European Space Agency (ESA) has provided free and open access to Synthetic Aperture Radar (SAR) data since 2014. This has led to a rapid increase in the availability of SAR data and has enabled a wide range of applications in Earth and environmental sciences. However, processing InSAR data has remained a challenge for non-experts. Advanced time series algorithms which combine persistent scatterer (PS) and distributed scatterer (DS) (also known as phase linking) have been developed for over a decade (Guarnieri & Tebaldini, 2008); however, existing open source tools have not included these techniques until recently.

The first prototype was the `FRInGE` C++ library (Fattahi et al., 2019), which implements

algorithms and workflows from (Ferretti et al., 2011) and (Ansari et al., 2018). The `Miaplpy` Python library contains a superset of the features in `FRInGE`, as well as new algorithms developed in (Mirzaee et al., 2023). The MATLAB `TomoSAR` library was also made public in 2022, which implements the "Compressed SAR" (ComSAR) algorithm- a variant of phase linking detailed in (Ho Tong Minh & Ngo, 2022).

`dolphin` has been developed for the Observational Products for End-Users from Remote Sensing Analysis (OPERA) project. OPERA is a Jet Propulsion Laboratory project funded by the Satellite Needs Working Group (SNWG) tasked with generating a North American Surface Displacement product. This product is required to cover over 10 million square kilometers of land at 30 meter resolution or finer with under 72 hours of latency; as such, new software was required which could handle the heavy computational demands of the project.

Overview of Dolphin

`dolphin` processes stacks of coregistered single-look complex (SLC) radar images into a time series of surface displacement. The software has pre-made workflows accessible through command line tools which call core algorithms for PS/DS processing:

- The `shp` subpackage estimates the SAR backscatter distribution to find neighborhoods of statistically homogeneous pixels (SHPs) using the generalized likelihood ratio test from Parizzi & Brcic (2011) or the Kolmogorov-Smirnov test from Ferretti et al. (2011).
- The `phase_link` subpackage processes the complex SAR covariance matrix into a time series of wrapped phase using the CAESAR algorithm (Fornaro et al., 2015), the eigenvalue-based maximum likelihood estimator of interferometric phase (EMI) (Ansari et al., 2018), or the combined phase linking (CPL) approach from (Mirzaee et al., 2023).
- The `unwrap` subpackage exposes multiple phase unwrapping algorithms, including the Statistical-cost, Network-flow Algorithm for Phase Unwrapping (SNAPHU) (Chen & Zebker, 2001) and the PHASS algorithm (available in the InSAR Scientific Computing Environment (Rosen et al., 2018)).
- The `timeseries` module contains basic functionality to invert an overdetermined network of unwrapped interferograms into a time series and estimate the average surface velocity.

The outputs of `dolphin` are also compatible with the Miami INsar Time-series software for users who are already comfortable with MintPy (Yunjun et al., 2019).

While `dolphin` has been primarily tested on CPU-based workstations and cloud environments, the compute-intensive routines leverage the accelerated linear algebra (XLA) compiler within `jax` library (Bradbury et al., 2018). This means that with only an extra installation, users can accelerate their processing by 5-20x on machines with a compatible GPU.

The `dolphin` command line tool provides an interface for running the end-to-end displacement workflow on large datasets (e.g. hundreds of full-frame Sentinel-1 images) with minimal required configuration. For example, if a user has created a stack of coregistered SLCs in a `data/` directory, they only need to follow two steps to run the full workflow with all default parameters:

1. Configure the workflow with the `config` command, indicating the location of the SLCs, which dumps the output to a YAML file:

```
dolphin config --slc-files data/*
```

2. Run the workflow saved in the YAML configuration file with the `run` command:

```
dolphin run dolphin_config.yaml
```

Figure 1 shows an example result of the final average surface velocity map created by `dolphin`. The inputs were OPERA Coregistered Single-Look Complex (CSLC) geocoded images from Sentinel-1 data between February 2017 - December 2020 over the Mojave Desert.

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